Log Book Entry 2.2

NB2420 Electronic Instrumentation · Manual 2.2 · Timed acquisitions

Hand in via BrightSpace

Name(s): Partner:

Date: Group:

PREPARATION

complete before the practicum

1. *What theories, laws, and/or formulas are most important for this experiment?*
2. *What do you expect to learn?*
3. *What do you expect to observe, and why?*

TIMING ACCURACY

Tasks A1, I1, C1

A 5 s measurement at a 50 ms target interval. Predict as homework, then fill the measured column from your own timing data.

Parameter	Predicted	Measured	Unit
Total measurement duration			s
Sampling interval (target)			ms
Minimum measured interval			ms
Maximum measured interval			ms
Average measured interval			ms
Jitter (max − min)			ms

Timing function used (`ticks_ms` or `ticks_us`), and why:

Maximum timing error as a % of the 50 ms target:

SAMPLING RATE LIMITS

Tasks A2, I2, I3, C2

Signal type	Test f [Hz]	Minimum required rate [Hz]	Achieved rate [Hz]	Sufficient?
Potentiometer (slow)	< 1			
Sine wave	50			
Sine wave	100			
Sine wave	300			

At which frequency do the fine details of the sine start to disappear, and what is the highest frequency you would trust?

SWEEP AND CONCLUSIONS

Tasks A4, I3, C3

Frequency sweep 1 Hz to 500 Hz, 200 samples per point

Peak voltage constant across the sweep?

Frequencies with the largest deviation:

Was 0.5 s settling enough?

How would you make this sweep more efficient without losing data quality?

Graph *Plot voltage against time for the potentiometer run, or measured amplitude against frequency for the sweep.*

REFLECTION

complete before submission

1. *Which results matched or differed from your expectations, and why?*
2. *What is the most important thing you learned?*
3. *What would you do differently next time?*

TA check-off: Date: